

Algebra PhD Exam, September 2000

Work exactly seven out of the following eleven exercises.

1. This problem has three parts.
 - (a) Define solvable group.
 - (b) Prove that the homomorphic image of a solvable group is solvable.
 - (c) Prove that a free group is solvable if and only if it is the free group on at most one generator.
2. Let G be a group; call $g \in G$ a non-generator if, for each subset X of G so that $X \cup \{g\}$ generates G , then, in fact, X itself generates G . Let $\text{Fr}(G)$ denote the set of all non-generators of G .
 - (a) Prove that $\text{Fr}(G)$ is a subgroup of G .
 - (b) Show that $\text{Fr}(G)$ is the intersection of all maximal (proper) subgroups of G . (Careful with Zorn's Lemma!)
3. Suppose that R is a principal ideal domain. Prove that any submodule of a free R -module is free.
4. Suppose $(m, n) = 1$. Compute $\mathbb{Z}_m \otimes_{\mathbb{Z}} \mathbb{Z}_n$. Justify your answer.
5. Let R be a finite semisimple ring with identity. Suppose that no fourth power $k^4 > 1$ ($k \in \mathbb{N}$) divides $|R|$. Prove that R is commutative, and therefore a direct product of fields.
6. Suppose that R is a ring with identity. Prove that

$$\text{Hom}_{\text{Ab}} \left(B, \prod_{i \in I} G_i \right) \cong \prod_{i \in I} \text{Hom}_{\text{Ab}}(B, G_i),$$

as right R -modules, for all left R -modules B and all abelian groups G_i ($i \in I$). Ab denotes the category of all abelian groups together with all homomorphisms between them.

You may use resources from category theory; if so, outline your argument so that it is clear which theorems you are appealing to.

7. Prove Nakayama's Lemma: let A be a commutative ring with identity. Let M be a finitely generated A -module, and I be an ideal of A , contained in the Jacobson radical $J(A)$ of A . Show that if $IM = M$ then $M = \{0\}$.
8. Let A be a commutative ring with identity. For each multiplicative system S of A , prove that $S^{-1}A$ is a flat A -module.
9. Suppose that A is a subring of the commutative ring B with 1. If B is integral over A , prove that every homomorphism f of A into the algebraically closed field L admits an extension to a homomorphism $g : B \rightarrow L$.
10. Let Θ_p be the p -th cyclotomic polynomial over the field $\mathbb{Q}$, where p is a prime number. Show that the Galois group of the splitting field of Θ_p over $\mathbb{Q}$ is cyclic of order $p - 1$. (Be clear about any theorems you quote.)
11. Consider the polynomial over the field $\mathbb{F}_2$ of two elements: $g(x) = x^4 + x^3 + x^2 + x + 1$.

- (a) Prove that $g(x)$ is irreducible over $\mathbb{F}_2$.
- (b) Let K be a splitting field for $g(x)$ over $\mathbb{F}_2$, and let $r \in K$ be a root of $g(x)$. Factor $g(x)$ into irreducibles over $\mathbb{F}_2(r)$.
- (c) Show that $K = \mathbb{F}_2(r)$.
- (d) Find the Galois group $\text{Gal}(K/\mathbb{F}_2)$.